E-Commerce Management System – Relational Schema

1. CUSTOMER

Attribute	Data Type	Constraint
Customer_id	INT	PRIMARY KEY, AUTO_INCREMENT
Name	VARCHAR(50)	NOT NULL
Email	VARCHAR(100)	NOT NULL, UNIQUE
Password	VARCHAR(250)	NOT NULL
Phone	VARCHAR(15)	NOT NULL, UNIQUE
Address	VARCHAR(250)	--

2. SELLER

Attribute	Data Type	Constraint
Seller_id	INT	PRIMARY KEY, AUTO_INCREMENT
Name	VARCHAR(50)	NOT NULL
Email	VARCHAR(100)	NOT NULL, UNIQUE
Password	VARCHAR(250)	NOT NULL
Phone	VARCHAR(15)	NOT NULL, UNIQUE
Address	VARCHAR(250)	--

3. ADMIN

Attribute	Data Type	Constraint
Admin_id	INT	PRIMARY KEY, AUTO_INCREMENT
Name	VARCHAR(50)	NOT NULL
Username	VARCHAR(20)	NOT NULL, UNIQUE
Password	VARCHAR(250)	NOT NULL

4. CATEGORY

Attribute	Data Type	Constraint
Category_id	INT	PRIMARY KEY, AUTO_INCREMENT
Category_name	VARCHAR(50)	NOT NULL, UNIQUE
Description	VARCHAR(250)	--

5. PRODUCT

Attribute	Data Type	Constraint
Product_id	INT	PRIMARY KEY, AUTO_INCREMENT
Seller_id	INT	FOREIGN KEY REFERENCES SELLER(Seller_id), NOT NULL
Category_id	INT	FOREIGN KEY REFERENCES CATEGORY(Category_id), NOT NULL
Product_name	VARCHAR(100)	NOT NULL
Description	TEXT	--
Price	DECIMAL(10,2)	NOT NULL, CHECK (Price >= 0)
Stock	INT	NOT NULL, CHECK (Stock >= 0)
Status	VARCHAR(20)	NOT NULL, DEFAULT 'Available', CHECK (Status IN ('Available', 'Out of Stock', 'Discontinued'))

6. CART

Attribute	Data Type	Constraint
Cart_id	INT	PRIMARY KEY, AUTO_INCREMENT
Customer_id	INT	FOREIGN KEY REFERENCES CUSTOMER(Customer_id) ON DELETE CASCADE, NOT NULL, UNIQUE
Created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

7. CART_ITEM

Attribute	Data Type	Constraint
Cart_item_id	INT	PRIMARY KEY, AUTO_INCREMENT

Cart_id	INT	FOREIGN KEY REFERENCES CART(Cart_id) ON DELETE CASCADE, NOT NULL
Product_id	INT	FOREIGN KEY REFERENCES PRODUCT(Product_id) ON DELETE CASCADE, NOT NULL
Quantity	INT	NOT NULL, CHECK (quantity > 0)
—	—	UNIQUE (cart_id, product_id)

8. ORDERS

Attribute	Data Type	Constraint
Order_id	INT	PRIMARY KEY, AUTO_INCREMENT
Customer_id	INT	FOREIGN KEY REFERENCES CUSTOMER(Customer_id), NOT NULL
Order_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
Total_amount	DECIMAL(10,2)	NOT NULL, CHECK (Total_amount >= 0)
Payment_method	VARCHAR(20)	NOT NULL, CHECK (Payment_method IN (‘COD’, ‘UPI’, ‘Card’))
Status	VARCHAR(20)	NOT NULL, DEFAULT 'Placed', CHECK (Status IN (‘Placed’, ’Confirmed’, ’Shipped’, ‘Delivered’, ‘Cancelled’))
Delivery_address	VARCHAR(250)	NOT NULL

9. ORDER_ITEM

Attribute	Data Type	Constraint
Order_item_id	INT	PRIMARY KEY, AUTO_INCREMENT
Order_id	INT	FOREIGN KEY REFERENCES ORDERS(Order_id) ON DELETE CASCADE, NOT NULL
Product_id	INT	FOREIGN KEY REFERENCES PRODUCT(Product_id) ON DELETE RESTRICT, NOT NULL
Quantity	INT	NOT NULL, CHECK (Quantity > 0)
Price	DECIMAL(10,2)	NOT NULL, CHECK (Price >= 0)

10. REVIEW

Attribute	Data Type	Constraint
Review_id	INT	PRIMARY KEY, AUTO_INCREMENT

Customer_id	INT	FOREIGN KEY REFERENCES CUSTOMER(Customer_id), NOT NULL
Product_id	INT	FOREIGN KEY REFERENCES PRODUCT(Product_id) ON DELETE CASCADE, NOT NULL
Rating	INT	NOT NULL, CHECK (Rating BETWEEN 1 AND 5)
Comment	VARCHAR(500)	--
Review_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
—	—	UNIQUE (customer_id, product_id)

Foreign Key Relationships

Foreign Key	References	Relationship
PRODUCT.Seller_id	SELLER.Seller_id	One Seller → Many Products
PRODUCT.Category_id	CATEGORY.Category_id	One Category → Many Products
CART.Customer_id	CUSTOMER.Customer_id	One Customer → One Cart
CART_ITEM.Cart_id	CART.Cart_id	One Cart → Many Cart Items
CART_ITEM.Product_id	PRODUCT.Product_id	One Product → Many Cart Items
ORDERS.Customer_id	CUSTOMER.Customer_id	One Customer → Many Orders
ORDER_ITEM.Order_id	ORDERS.Order_id	One Order → Many Order Items
ORDER_ITEM.Product_id	PRODUCT.Product_id	One Product → Many Order Items
REVIEW.Customer_id	CUSTOMER.Customer_id	One Customer → Many Reviews
REVIEW.Product_id	PRODUCT.Product_id	One Product → Many Reviews

Additional Constraints / Design Notes

- In CART, Customer_id is UNIQUE so each customer has one cart.
- CART_ITEM has UNIQUE(Cart_id, Product_id) so the same product is not stored twice in one cart.
- In ORDER_ITEM, Price stores the product price at the time of purchase, so later product-price changes do not alter old order records.
- In REVIEW, Rating is restricted to values from 1 to 5.
- REVIEW has UNIQUE(customer_id, product_id) so a customer cannot submit more than one review per product.
- In PRODUCT, Price and Stock are restricted to non-negative values; In ORDER_ITEM, Quantity and In CART_ITEM, Quantity are restricted to positive values.
- In Order, Status can use values such as Placed, Confirmed, Shipped, Delivered and Cancelled.